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AI CAREER FOR WOMEN

Science



StudentPulse: A Student Learning Performance and Study Planning Dashboard

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide and faculty for their guidance and encouragement during the development of StudentPulse. We also thank our institution for providing an opportunity to apply our learning through this project. Finally, we appreciate the support of everyone who contributed ideas, feedback, and motivation. (Please personalise this acknowledgement and add the guide/institution names before submission.)

StudentPulse is a browser-based student learning dashboard designed to help learners reflect on study habits and turn them into practical study actions. Students can enter study hours, attendance, previous exam score, assignment completion, sleep duration, class participation, weekly revision sessions, and a target score. The site calculates an illustrative performance estimate and presents suggested next steps and what-if comparisons. It also includes a weekly study checklist, a focus timer, a subject tracker, auto-saved study notes, daily learning check-ins, and prediction-history export. Check-ins and estimates are described by the website as being saved locally in the browser, with CSV/report export options. The project follows an interactive web-application approach, using user-provided inputs to update a learning snapshot and support planning. The live site explicitly states that its scoring formula is illustrative, not a trained or validated machine-learning model, and should not be used for grading, admissions, or other consequential decisions. StudentPulse therefore serves as a self-reflection and study-organisation aid rather than an authoritative predictor of academic outcomes.

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CHAPTER 1

Introduction

1.1 Problem Statement: Students may find it difficult to connect everyday study habits—such as attendance, revision, sleep, and assignment completion—with a clear plan for improvement. Information is often spread across separate tools, while a score alone does not explain what action to take next. StudentPulse addresses this organisational gap by bringing a learning profile, illustrative estimate, study suggestions, and planning tools into one browser-based workspace.

1.2 Motivation: The project was chosen to explore how a simple, interactive web application can encourage consistent study habits and self-reflection. A single workspace for estimating, planning, tracking subjects, recording notes, and checking in may help students identify small actions they can take toward their own learning goals.

1.3 Objectives: (i) provide an interactive learning profile using configurable study-related inputs; (ii) display an illustrative performance estimate and target gap; (iii) suggest practical next steps and compare hypothetical scenarios; (iv) provide a weekly checklist and focus timer; (v) track subjects, notes, and daily check-ins; and (vi) allow users to review or export locally saved history.

1.4 Scope of the Project: StudentPulse is a browser-based student self-management and reflection tool. Its scope includes input-based illustrative estimates, study planning, focus sessions, subject tracking, notes, daily check-ins, and local history/export features. The website does not establish that its estimate is scientifically validated, does not demonstrate a trained predictive model, and should not be used for grading, admissions, or consequential decisions. The visible website does not provide evidence of a server-side account system or institution-wide analytics.

CHAPTER 2

Literature Survey

2.1 Review of Related Work: Student learning dashboards commonly present indicators, progress summaries, and planning tools to help learners reflect on their activities. Study-planning tools also use checklists, timers, and revision routines to encourage consistent work. StudentPulse combines these familiar patterns in one lightweight interface. The live site describes its estimate as illustrative, so it should be understood as a planning aid rather than evidence of predictive accuracy.

2.2 Related Techniques: The project uses an input-driven scoring approach, interactive form controls, scenario comparison, checklist-based planning, a focus timer, subject tracking, text notes, daily check-ins, and browser-local persistence/export as described by the website. The public interface does not identify a trained machine-learning algorithm or provide a validated statistical model.

2.3 Gap and Project Approach: A score by itself may not guide a student toward a concrete next step. StudentPulse addresses this usability gap by combining the estimate with suggested actions and tools for planning and reflection. However, the current illustrative formula has no published validation results on the website; any future predictive use would require suitable data, evaluation, fairness checks, and privacy safeguards.

CHAPTER 3

Proposed Methodology

3.1 System Design: The application is organised into a dashboard and supporting workspaces: study planner, focus lab, prediction history, and project overview.

3.1.1 User Setup: The public website presents editable learning-profile inputs and does not show a mandatory registration workflow. Users can adjust the profile directly in the browser.

3.1.2 Estimate Calculation: When users submit their profile, the interface displays an illustrative performance estimate, target-score distance, suggested next steps, and what-if comparisons. The formula is explicitly described as illustrative, not validated.

3.2 Modules Used: The visible application includes the following functional modules.

3.2.1 Learning Profile and Estimate: Inputs include study hours per day, attendance, previous exam score, assignment completion, sleep per night, class participation, weekly revision sessions, and target score.

3.2.2 Study Planner: A weekly study plan presented as a checklist. Tasks are ticked off as they are completed, a percentage-complete indicator is shown, progress is saved on the device, and the checklist can be reset. The page also lists habits that tend to help: active recall, spaced revision and focused breaks.

3.2.3 Focus Lab: A personal study space with a focus timer (default 25:00; focus options of 15, 25, 45 or 50 minutes and break options of 5, 10 or 15 minutes), a subject tracker for subjects or projects, auto-saved study notes (up to 5000 characters) and a daily check-in.

3.2.4 Prediction History and Export: Lists the estimates and check-ins saved in the browser, with an Export CSV option and an Export report button on the dashboard. The site states that the history stays in the browser unless cleared and is not sent to a server.

3.2.5 About the Project: A project overview describing interactive estimates, the study planner and private local history, together with an Important limitation notice that the scoring formula is illustrative, not a trained or validated machine-learning model.

Table 2: Functional modules of StudentPulse

Module	Purpose (as shown on the website)
Dashboard / Learning profile	Eight adjustable inputs, Calculate estimate, score estimate, target gap, next steps, what-if comparison
Study planner	Weekly checklist with percentage complete and reset; study habits that tend to help
Focus lab	Focus timer, subject tracker, study notes, daily check-in, focus sessions completed
Prediction history	Recent check-ins saved locally; Export CSV; clear history
About the project	Project overview and important limitation notice

3.3 Data Flow Description

At a high level, the user enters learning-profile values; the application processes those values into an illustrative estimate and suggestions; the dashboard displays the results; and supported history, notes, check-ins, and checklist progress are stored locally in the browser according to the live site. This description is based on the visible interface and does not imply a backend database.

3.3.1. Level 0: Student → StudentPulse web interface → illustrative estimate, suggested actions, study plan, and local records.

3.3.2. Level 1 – Profile and Estimate: User enters or adjusts profile values → selects Calculate estimate → application calculates an illustrative score → dashboard presents score, target gap, and suggestions.

3.3.3. Level 1 – Study Tools: User manages weekly checklist, focus timer, subjects, notes, and daily check-in → interface updates progress and stores supported information locally.

3.3.4. Level 1 – History and Export: Saved estimates/check-ins are shown in prediction history → user can export history as CSV or export a report. Data is described as staying in the current browser unless cleared.

3.4 Advantages: One workspace for study reflection and planning; configurable inputs; immediate interactive feedback; weekly checklist; focus timer; subject tracker; study notes; daily check-ins; and locally stored history with export options. The tools are designed to support planning rather than make high-stakes decisions.

3.5 Requirement Specification

3.5.1. Hardware Requirements: A desktop, laptop, tablet, or smartphone capable of running a modern web browser; a display and input device; and an internet connection to access the hosted website. Specific minimum hardware specifications are not stated on the site.

3.5.2 Software Requirements: A modern web browser and access to https://studentpulsee.netlify.app. The public interface does not disclose a complete development stack, package list, or backend details, so these are not asserted here.

Table 3: Hardware and software requirements

Item	Requirement
Device	Desktop, laptop, tablet or smartphone able to run a modern web browser
Connectivity	Internet connection to open the hosted website (https://studentpulsee.netlify.app)
Software	Modern web browser; the site stores data in the browser on the same device
Not stated on the site	Minimum hardware specifications, development stack and backend details

CHAPTER 4

Implementation and Results

4.1 Results of Learning Profile and Estimate: The dashboard exposes inputs for study hours, attendance, previous exam score, assignment completion, sleep, class participation, revision sessions, and target score. It provides a Calculate estimate action and an estimate panel. The site labels the score as illustrative. No independent test dataset, accuracy metrics, or validation study is published on the page.

The dashboard opens with four summary cards: Latest estimate, Study hours per day, Attendance and Saved check-ins. A side menu gives access to Dashboard, Study planner, Focus lab, Prediction history and About the project, and the header offers Appearance and Export report controls. A card on the side menu reminds the student that small steps add up and that effort matters more than one score. Figure 1 shows the dashboard captured from the live website.

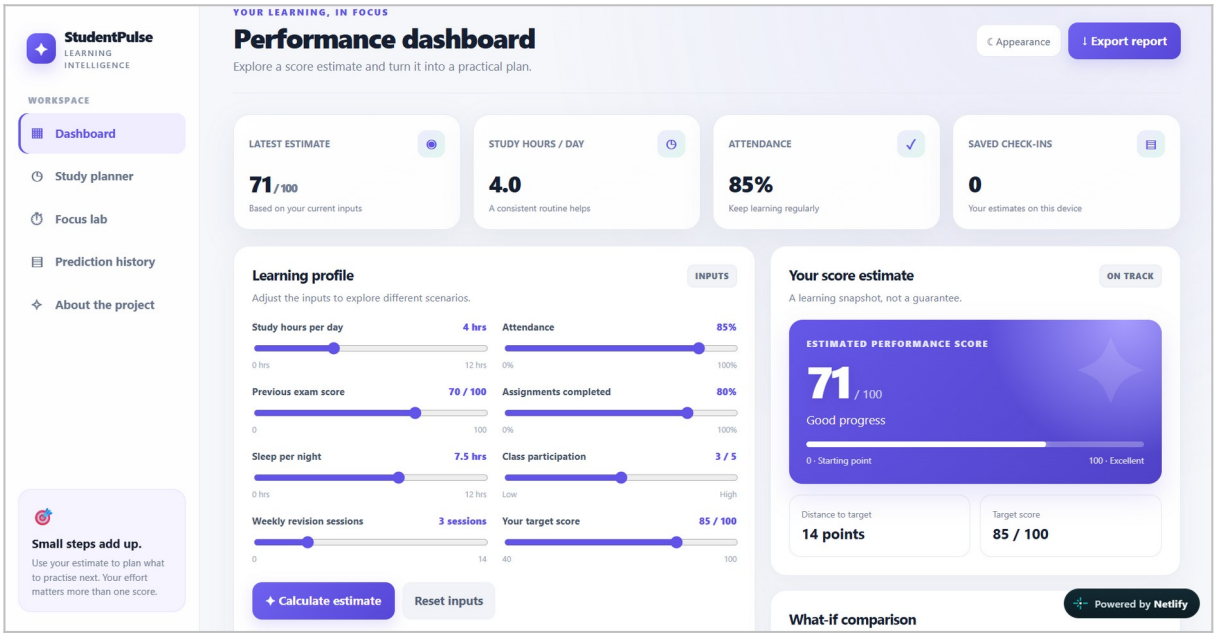


Figure 1: Performance dashboard with summary cards, learning profile and score estimate

Table 1: Learning-profile inputs on the dashboard

Input	Range	Default shown
Study hours per day	0 – 12 hrs	4 hrs
Attendance	0 – 100%	85%
Previous exam score	0 – 100	70 / 100
Assignments completed	0 – 100%	80%
Sleep per night	0 – 12 hrs	7.5 hrs
Class participation	Low – High (1 – 5)	3 / 5
Weekly revision sessions	0 – 14	3 sessions
Your target score	40 – 100	85 / 100

The Learning profile panel (Figure 2) holds the eight sliders listed in Table 1, with Calculate estimate and Reset inputs buttons. The Your score estimate panel (Figure 3) shows the estimated performance score out of 100 on a scale from 0 (starting point) to 100 (excellent), the distance to the target and the target score. In the captured screenshot the default inputs gave an estimate of 71 / 100, labelled “Good progress”, 14 points from the 85 / 100 target, with the panel tagged “On track” and the note “A learning snapshot, not a guarantee.” The site describes this value as an illustrative estimate, not a validated prediction.

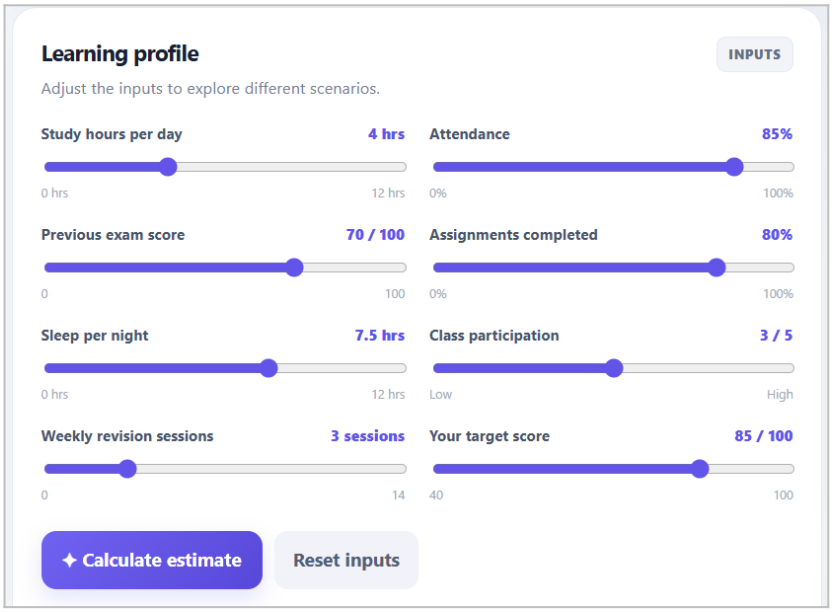


Figure 2: Learning profile inputs

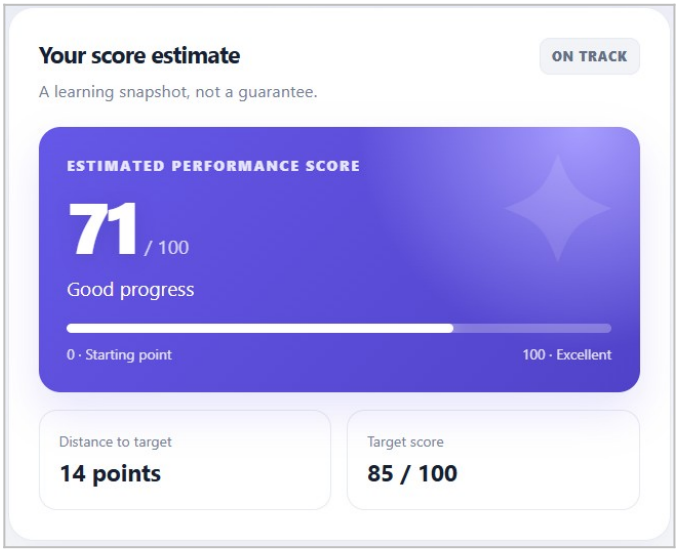


Figure 3: Performance estimate panel

4.2 Results of Study Planning Tools: The website includes a weekly study checklist, subject tracker, notes area, and daily check-in. The interface indicates that checklist progress and personal records are saved on the current device/browser. These features support organisation and reflection; their educational impact has not been quantified in the public interface.

The Study planner page, "Your weekly study plan", lists one suggested task for each day of the week: a 25-minute focus session (Monday), an assignment block (Tuesday), active recall (Wednesday), ask for clarity (Thursday), practice questions (Friday), spaced revision (Saturday) and weekly reflection (Sunday). Each task has a tick box, a percentage-complete badge shows progress (0% complete in Figure 4), progress is saved on the device, and a Reset weekly checklist button clears it.

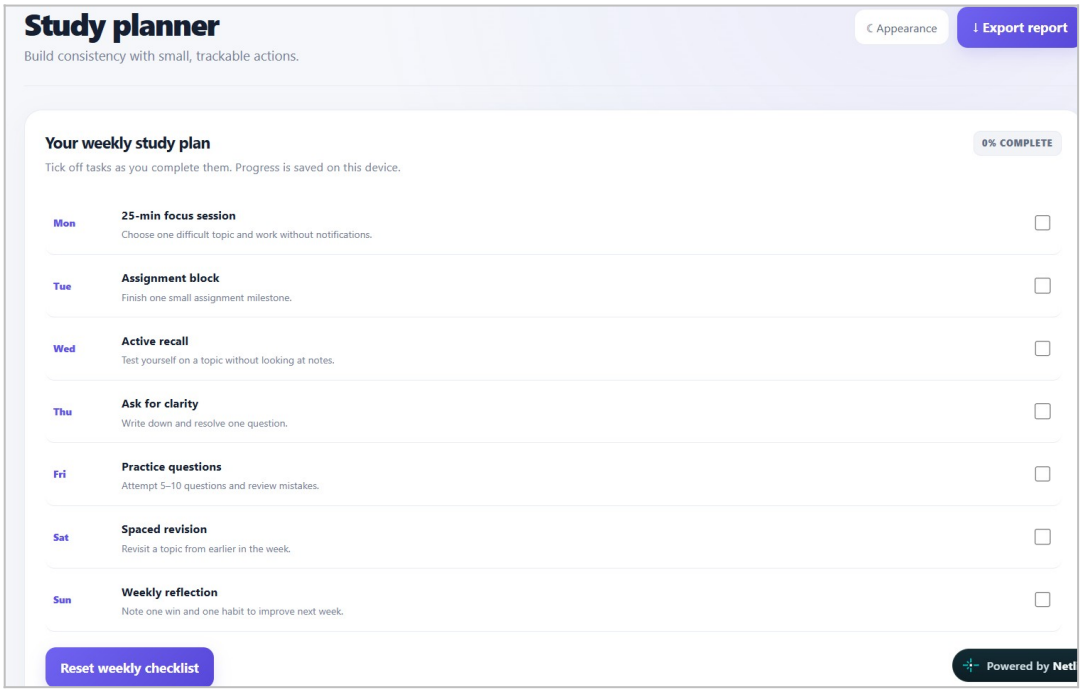


Figure 4: Weekly study plan (Study planner)

4.3 Results of Focus and History Features: The Focus Lab provides a focus timer with configurable focus and break durations, subject tracking, notes, and daily check-ins. Prediction history can be

reviewed and exported as CSV, and the dashboard includes an export-report control. The public site does not provide measured results showing improved concentration or grades.

The Focus lab (Figure 5) is described as a personal study space saved on the device. The Focus timer starts at 25:00 with Start and Reset buttons, a Focus minutes selector (25 minutes shown; options of 15, 25, 45 and 50 minutes) and a Break minutes selector (5 minutes shown; options of 5, 10 and 15 minutes), with a reminder to keep the tab open while the timer runs because completing a session records it locally. The Subject tracker has a field for a subject or project name (for example DBMS, React or Maths), a number field and an Add button, and shows “0 subjects” until one is added.

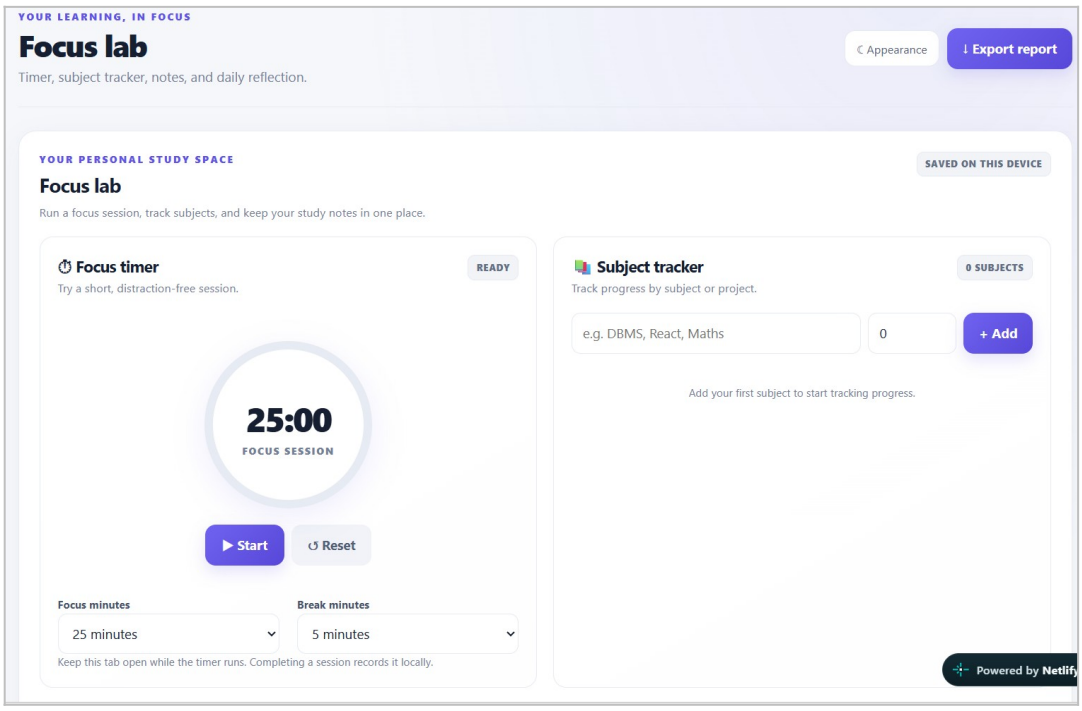


Figure 5: Focus timer and subject tracker (Focus lab)

Below the timer and tracker (Figure 6), Study notes provides an auto-saved text area (0 / 5000 characters shown) with a Clear notes button. The Daily check-in, labelled “No judgement”, asks “How focused do you feel today?” with four options (Low, Okay, Good, Great), a field for one small win and a Save daily check-in button, followed by a Focus sessions completed counter. Saved estimates and check-ins appear under Recent check-ins and in Prediction history, where they can be exported with Export CSV or cleared; the site states that this history stays in the browser and is not sent to a server.

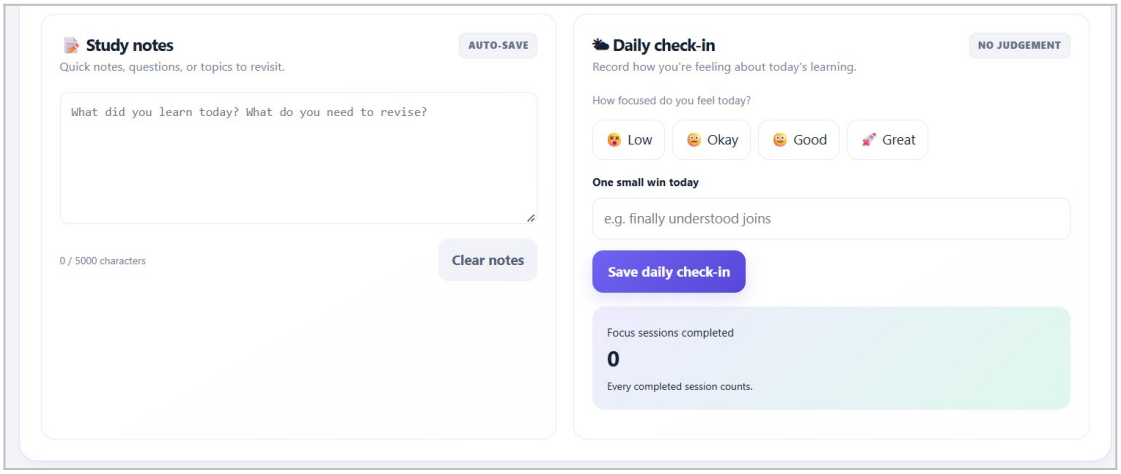


Figure 6: Study notes and daily check-in (Focus lab)

The About StudentPulse page (Figure 7) explains that the project turns study habits into an interactive learning snapshot and suggested actions, highlights interactive estimates, the study planner and private local history, and shows an Important limitation notice: the scoring formula is illustrative, is not a trained or validated machine-learning model and must not be used for grading, admissions or consequential decisions.

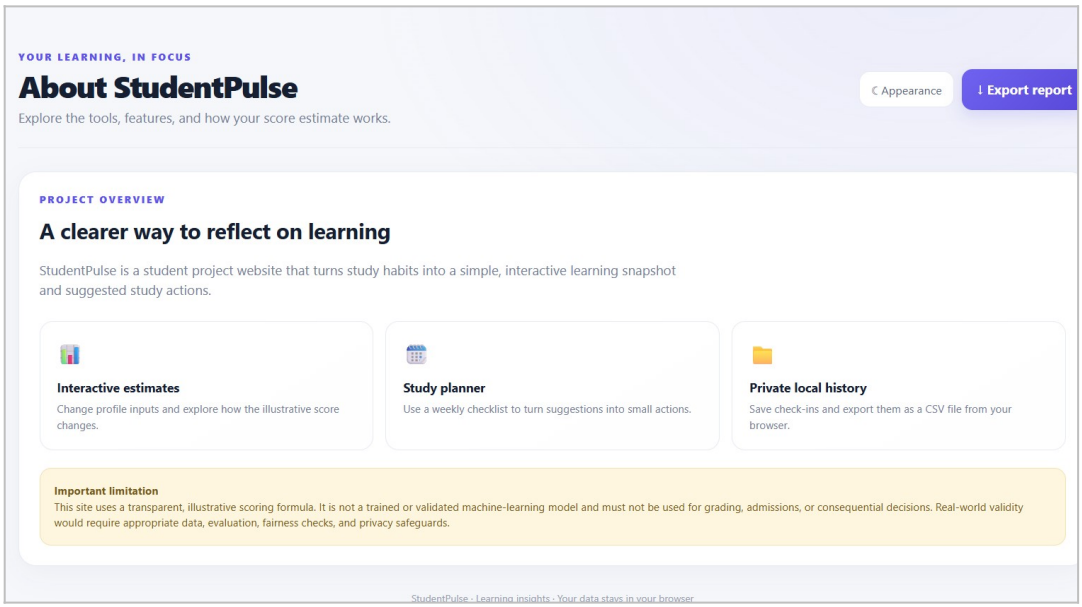


Figure 7: About the project page with limitation notice

CHAPTER 5

Discussion and Conclusion

5.1 **Key Findings:** StudentPulse brings together an illustrative learning estimate, suggested actions, what-if comparison, weekly planning, focus timing, subject tracking, notes, daily check-ins, and local history/export in one interface. Its main contribution is the integration of reflection and planning tools. The site does not provide evidence that the estimate is a

validated predictor or that the tools have been experimentally shown to improve academic performance.

5.2 **Project Link:** Live website: https://studentpulsee.netlify.app. GitHub repository: https://github.com/khyatiagarwal0751-ops/studentpulse

5.3 **Video Demonstration:** A demonstration-video link was not visible on the public website reviewed. Add the video URL here if one has been recorded.

5.4 **Limitations:** The score is based on a transparent illustrative formula and is not a trained or validated machine-learning model. The site warns against using it for grading, admissions, or consequential decisions. Publicly visible information does not establish predictive accuracy, fairness, or measured educational impact. Locally stored data may be browser/device-specific, and clearing browser data may remove it. The full implementation stack and backend architecture are not specified by the visible page.

5.5 **Future Work:** Future improvements could include usability testing with students, clearer explanation of the scoring formula, evaluation against appropriately collected and consented data before making predictive claims, fairness and privacy assessments, accessibility and mobile testing, and optional secure synchronisation if justified. Any model-based prediction should be validated and must not be used for consequential decisions without rigorous evidence and safeguards.

5.6 **Conclusion:** StudentPulse demonstrates how a browser-based dashboard can combine study reflection with practical planning tools. It enables users to explore their learning profile, receive an illustrative estimate, plan weekly tasks, run focus sessions, track subjects, write notes, and review locally saved history. The project is best positioned as a self-management aid. Its estimate is explicitly not validated, so further research and evaluation would be necessary before claiming reliable prediction of academic outcomes.

REFERENCES

[1]. [1]. StudentPulse, “StudentPulse | Performance Insights,” live project website. Available: https://studentpulsee.netlify.app (accessed 13 October 2026).

Appendices

Appendix A – Live Website: https://studentpulsee.netlify.app

Appendix B – Screenshots: Add dated screenshots of the dashboard, learning profile, estimate, weekly planner, focus lab, and prediction history.

Appendix C – Source Code: https://github.com/khyatiagarwal0751-ops/studentpulse